

Weekly Assessments Every Friday

16 Weeks | 2 Hours per Day

Python Full Stack Training

1. Program Snapshot

2. Month 1: Python
Fundamentals

3. Month 2: Frontend
Foundations

4. Month 3: JavaScript & Django
Fundamentals

5. Month 4: Project, Deployment
& Career Prep

6. Assessments & Deliverables

Program Snapshot

Outcome: Job-ready **Python Full Stack Developer** with a real-time Django project.

DURATION

4 Months
(16 Weeks)

CLASS TIMING

2 Hours
Per Day

ASSESSMENTS

Weekly
Every Friday (Weekly Assessment)

TECHNOLOGIES COVERED

Python (Basics to OOP)

HTML

CSS

JavaScript

Django

MySQL

CURRICULUM STRUCTURE

01

Python Fundamentals

WEEKS 1-4

Introduction, Control Structures, Data Structures, OOP.

02

Frontend Development

WEEKS 5-8

HTML5, CSS3, Layouts, Responsive Design.

03

JavaScript & Django

WEEKS 9-12

DOM Manipulation, Django MVT, Models, Forms.

04

Project & Career Prep

WEEKS 13-16

Capstone Project, Deployment, Interview Preparation.

Curriculum Structure



Python Programming Fundamentals

Focus: Building a strong foundation in Python syntax, control flow mechanisms, essential data structures, and Object-Oriented Programming (OOP) principles.

WEEK 1

Intro to Python & Basics

- Python installation & IDE setup
- Variables & data types
- Operators & Expressions
- Input & output handling

**WEEK 2**

Control Structures & Functions

- Conditionals (if, elif, else)
- Loops (for, while)
- Functions & arguments
- Logic building exercises

**WEEK 3**

Python Data Structures

- Strings & string methods
- Lists & Tuples
- Sets & Dictionaries
- List comprehensions

**WEEK 4**

Object-Oriented Programming

- Classes & Objects
- Inheritance & Polymorphism
- Encapsulation
- File Handling basics

Python Programming Fundamentals

WEEK	TOPICS COVERED	FRIDAY ASSESSMENT
1 Week 1	What is Python, applications, install & IDE setup, variables & data types, operators, input & output	Python basics MCQs + coding
2 Week 2	Conditional statements (if, elif, else), loops (for, while), break & continue, functions, arguments & return values	Logic building & function coding test
3 Week 3	Strings & string methods, lists & list methods, tuples, sets, dictionaries, list & dictionary comprehensions	Data structures practical
4 Week 4	Classes & objects, constructors, inheritance, polymorphism, encapsulation, exception handling, file handling basics	OOP & file handling test

Month 2: Frontend Development

Focusing on building the visual structure and interactivity of web applications through HTML, CSS, and introductory JavaScript.

WEEK 5**HTML Fundamentals**

- HTML structure & tags
- Lists, links, images
- Tables & formatting

WEEK 6**HTML Forms & Semantics**

- Forms & input types
- Validation basics
- Semantic elements

WEEK 7**CSS Basics & Styling**

- Selectors & Box model
- Fonts & colors
- Backgrounds & borders

WEEK 8**CSS Layouts & JS Basics**

- Flexbox & Responsive
- Media queries
- JS variables & logic

Frontend Development

Detailed curriculum breakdown for Weeks 5-8

WEEK	TOPICS COVERED	FRIDAY ASSESSMENT
Week 5	Web development intro, HTML structure, text formatting, lists/links/images, tables	 HTML page creation Practical Assessment
Week 6	Forms & input types, form validation basics, semantic HTML, best practices	—
Week 7	CSS intro, selectors, box model, fonts & colors, backgrounds & borders	 CSS styling task Practical Assessment
Week 8	Flexbox, responsive design, media queries, JavaScript intro, variables/data types, operators/conditions	 Responsive UI + JS basics Practical Assessment

Month 3: JavaScript & Django

Transitioning from client-side interactivity to server-side architecture. Students master DOM manipulation before building robust backend systems with Django.

WEEK 09

JavaScript & DOM

- Functions, Arrays & Objects
- DOM Manipulation
- Event Handling
- Form Validation

WEEK 10

Django Intro

- MVT Architecture
- Project & App Structure
- URL Routing
- Views & Responses

WEEK 11

Templates & Models

- Template Inheritance
- Models & ORM
- Database Config
- CRUD Operations

WEEK 12

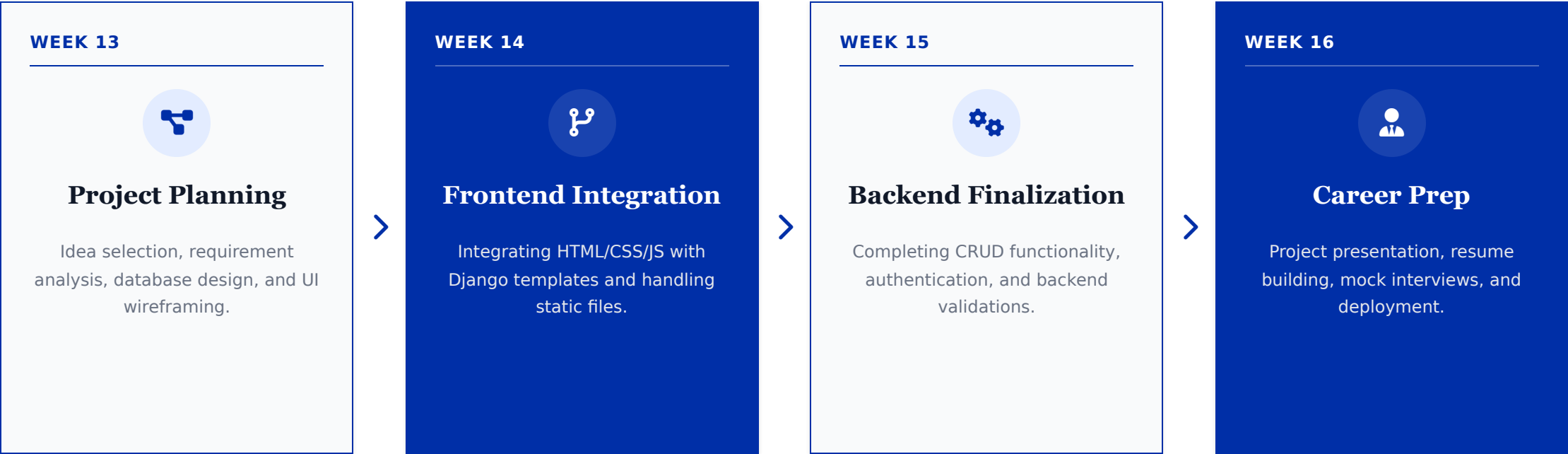
Forms & Auth

- Django Forms
- Model Forms
- User Authentication
- Login & Sessions

TIMELINE	TOPICS COVERED	FRIDAY ASSESSMENT
Week 9	JavaScript & DOM Functions & arrays, objects, DOM manipulation, events, form validation	 DOM-based mini project
Week 10	Django Fundamentals What is Django, MVT architecture, project & app structure, URL routing, views & HttpResponse	 Django setup & routing
Week 11	Templates & Models Templates, template inheritance, models & ORM, DB configuration, CRUD operations	 Django CRUD app
Week 12	Forms & Authentication Django forms, model forms, user authentication, login/logout, sessions	 Authentication-based app

Capstone Project & Career Preparation

A comprehensive final phase focusing on integrating full-stack skills into a real-world project and preparing for professional opportunities.



Project, Deployment & Interview Prep

The final phase consolidates learning through a capstone project and prepares students for the professional environment with deployment skills and interview practice.



Full Stack Django Project + Viva

Capstone Milestone

- 13

Project Planning
Project idea selection, requirement analysis, database design, UI wireframes
- 14

Frontend Integration
HTML/CSS/JS integration with Django, template rendering, static files handling
- 15

Backend Finalization
Complete CRUD functionality, authentication integration, basic validations
- 16

Project Presentation & Career Prep
Project deployment overview, resume preparation, interview questions, mock interviews

Student Deliverables Framework

WEEKLY

Weekly Assessments

Conducted every Friday to evaluate understanding of the week's topics.



ONGOING

Mini Projects

Practical coding assignments distributed throughout the program duration.



CAPSTONE

Final Project

One complete Django-based full stack application demonstrating comprehensive skills.



OPTIONAL

GitHub Repository

Maintenance of a code repository to showcase project history and version control skills.

